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The Development of Philosophical Ideas Since Descartes in Comparison with the New Situation in Quantum Theory

IN THE two thousand years that followed the culmination of Greek science and culture in the fifth and fourth centuries B.C. the human mind was to a large extent occupied with problems of a different kind from those of the early period. In the first centuries of Greek culture the strongest impulse had come from the immediate reality of the world in which we live and which we perceive by our senses. This reality was full of life and there was no good reason to stress the distinction between matter and mind or between body and soul. But in the philosophy of Plato one already sees that another reality begins to become stronger. In the famous simile of the cave Plato compares men to prisoners in a cave who are bound and can look in only one direction. They have a fire behind them and see on a wall the shadows of themselves and of objects behind them. Since they see nothing but the shadows, they regard those shadows as real and are not aware of the objects. Finally one of the prisoners escapes and comes from the cave into the light of the sun. For the first time he sees real things and realizes that he had been deceived

hitherto by the shadows. For the first time he knows the truth and thinks only with sorrow of his long life in the darkness. The real philosopher is the prisoner who has escaped from the cave into the light of truth, he is the one who possesses real knowledge. This immediate connection with truth or, we may in the Christian sense say, with God is the new reality that has begun to become stronger than the reality of the world as perceived by our senses. The immediate connection with God happens within the human soul, not in the world, and this was the problem that occupied human thought more than anything else in the two thousand years following Plato. In this period the eyes of the philosophers were directed toward the human soul and its relation to God, to the problems of ethics, and to the interpretation of the revelation but not to the outer world. It was only in the time of the Italian Renaissance that again a gradual change of the human mind could be seen, which resulted finally in a revival of the interest in nature.

The great development of natural science since the sixteenth and seventeenth centuries was preceded and accompanied by a development of philosophical ideas which were closely connected with the fundamental concepts of science. It may therefore be instructive to comment on these ideas from the position that has finally been reached by modern science in our time.

The first great philosopher of this new period of science was René Descartes who lived in the first half of the seventeenth century. Those of his ideas that are most important for the development of scientific thinking are contained in his *Discourse on Method*. On the basis of doubt and logical reasoning he tries to find a completely new and as he thinks solid ground for a philosophical system. He does not accept revelation as such a basis nor does he want to accept uncritically what is perceived

by the senses. So he starts with his method of doubt. He casts his doubt upon that which our senses tell us about the results of our reasoning and finally he arrives at his famous sentence: "cogito ergo sum." I cannot doubt my existence since it follows from the fact that I am thinking. After establishing the existence of the I in this way he proceeds to prove the existence of God essentially on the lines of scholastic philosophy. Finally the existence of the world follows from the fact that God had given me a strong inclination to believe in the existence of the world, and it is simply impossible that God should have deceived me.

This basis of the philosophy of Descartes is radically different from that of the ancient Greek philosophers. Here the starting point is not a fundamental principle or substance, but the attempt of a fundamental knowledge. And Descartes realizes that what we know about our mind is more certain than what we know about the outer world. But already his starting point with the "triangle" God-World-I simplifies in a dangerous way the basis for further reasoning. The division between matter and mind or between soul and body, which had started in Plato's philosophy, is now complete. God is separated both from the I and from the world. God in fact is raised so high above the world and men that He finally appears in the philosophy of Descartes only as a common point of reference that establishes the relation between the I and the world.

While ancient Greek philosophy had tried to find order in the infinite variety of things and events by looking for some fundamental unifying principle, Descartes tries to establish the order through some fundamental division. But the three parts which result from the division lose some of their essence when any one part is considered as separated from the other two parts. If one uses the fundamental concepts of Descartes at all, it is essential

that God is in the world and in the I and it is also essential that the I cannot be really separated from the world. Of course Descartes knew the undisputable necessity of the connection, but philosophy and natural science in the following period developed on the basis of the polarity between the "res cogitans" and the "res extensa," and natural science concentrated its interest on the "res extensa." The influence of the Cartesian division on human thought in the following centuries can hardly be overestimated, but it is just this division which we have to criticize later from the development of physics in our time.

Of course it would be wrong to say that Descartes, through his new method in philosophy, has given a new direction to human thought. What he actually did was to formulate for the first time a trend in human thinking that could already be seen during the Renaissance in Italy and in the Reformation. There was the revival of interest in mathematics which expressed an increasing influence of Platonic elements in philosophy, and the insistence on personal religion. The growing interest in mathematics favored a philosophical system that started from logical reasoning and tried by this method to arrive at some truth that was as certain as a mathematical conclusion. The insistence on personal religion separated the I and its relation to God from the world. The interest in the combination of empirical knowledge with mathematics as seen in the work of Galileo was perhaps partly due to the possibility of arriving in this way at some knowledge that could be kept apart completely from the theological disputes raised by the Reformation. This empirical knowledge could be formulated without speaking about God or about ourselves and favored the separation of the three fundamental concepts God-World-I or the separation between "res cogitans" and "res extensa." In this period there was in some

cases an explicit agreement among the pioneers of empirical science that in their discussions the name of God or a fundamental cause should not be mentioned.

On the other hand, the difficulties of the separation could be clearly seen from the beginning. In the distinction, for instance, between the "res cogitans" and the "res extensa" Descartes was forced to put the animals entirely on the side of the "res extensa." Therefore, the animals and the plants were not essentially different from machines, their behavior was completely determined by material causes. But it has always seemed difficult to deny completely the existence of some kind of soul in the animals, and it seems to us that the older concept of soul for instance in the philosophy of Thomas Aquinas was more natural and less forced than the Cartesian concept of the "res cogitans," even if we are convinced that the laws of physics and chemistry are strictly valid in living organisms. One of the later consequences of this view of Descartes was that, if animals were simply considered as machines, it was difficult not to think the same about men. Since, on the other hand, the "res cogitans" and the "res extensa" were taken as completely different in their essence, it did not seem possible that they could act upon each other. Therefore, in order to preserve complete parallelism between the experiences of the mind and of the body, the mind also was in its activities completely determined by laws which corresponded to the laws of physics and chemistry. Here the question of the possibility of "free will" arose. Obviously this whole description is somewhat artificial and shows the grave defects of the Cartesian partition.

On the other hand in natural science the partition was for several centuries extremely successful. The mechanics of Newton and all the other parts of classical physics constructed after its

model started from the assumption that one can describe the world without speaking about God or ourselves. This possibility soon seemed almost a necessary condition for natural science in general.

But at this point the situation changed to some extent through quantum theory and therefore we may now come to a comparison of Descartes's philosophical system with our present situation in modern physics. It has been pointed out before that in the Copenhagen interpretation of quantum theory we can indeed proceed without mentioning ourselves as individuals, but we cannot disregard the fact that natural science is formed by men. Natural science does not simply describe and explain nature; it is a part of the interplay between nature and ourselves; it describes nature as exposed to our method of questioning. This was a possibility of which Descartes could not have thought, but it makes the sharp separation between the world and the I impossible.

If one follows the great difficulty which even eminent scientists like Einstein had in understanding and accepting the Copenhagen interpretation of quantum theory, one can trace the roots of this difficulty to the Cartesian partition. This partition has penetrated deeply into the human mind during the three centuries following Descartes and it will take a long time for it to be replaced by a really different attitude toward the problem of reality.

The position to which the Cartesian partition has led with respect to the "res extensa" was what one may call metaphysical realism. The world, i.e., the extended things, "exist." This is to be distinguished from practical realism, and the different forms of realism may be described as follows: We "objectivate" a statement if we claim that its content does not depend on the

conditions under which it can be verified. Practical realism assumes that there are statements that can be objectivated and that in fact the largest part of our experience in daily life consists of such statements. Dogmatic realism claims that there are no statements concerning the material world that cannot be objectivated. Practical realism has always been and will always be an essential part of natural science. Dogmatic realism, however, is, as we see it now, not a necessary condition for natural science. But it has in the past played a very important role in the development of science; actually the position of classical physics is that of dogmatic realism. It is only through quantum theory that we have learned that exact science is possible without the basis of dogmatic realism. When Einstein has criticized quantum theory he has done so from the basis of dogmatic realism. This is a very natural attitude. Every scientist who does research work feels that he is looking for something that is objectively true. His statements are not meant to depend upon the conditions under which they can be verified. Especially in physics the fact that we can explain nature by simple mathematical laws tells us that here we have met some genuine feature of reality, not something that we have—in any meaning of the word—invented ourselves. This is the situation which Einstein had in mind when he took dogmatic realism as the basis for natural science. But quantum theory is in itself an example for the possibility of explaining nature by means of simple mathematical laws without this basis. These laws may perhaps not seem quite simple when one compares them with Newtonian mechanics. But, judging from the enormous complexity of the phenomena which are to be explained (for instance, the line spectra of complicated atoms), the mathematical scheme of quantum theory is comparatively

simple. Natural science is actually possible without the basis of dogmatic realism.

Metaphysical realism goes one step further than dogmatic realism by saying that "the things really exist." This is in fact what Descartes tried to prove by the argument that "God cannot have deceived us." The statement that the things really exist is different from the statement of dogmatic realism in so far as here the word "exists" occurs, which is also meant in the other statement "cogito ergo sum" . . . "I think, therefore I am." But it is difficult to see what is meant at this point that is not yet contained in the thesis of dogmatic realism; and this leads us to a general criticism of the statement "cogito ergo sum," which Descartes considered as the solid ground on which he could build his system. It is in fact true that this statement has the certainty of a mathematical conclusion, if the words "cogito" and "sum" are defined in the usual way or, to put it more cautiously and at the same time more critically, if the words are so defined that the statement follows. But this does not tell us anything about how far we can use the concepts of "thinking" and "being" in finding our way. It is finally in a very general sense always an empirical question how far our concepts can be applied.

The difficulty of metaphysical realism was felt soon after Descartes and became the starting point for the empiristic philosophy, for sensualism and positivism.

The three philosophers who can be taken as representatives for early empiristic philosophy are Locke, Berkeley and Hume. Locke holds, contrary to Descartes, that all knowledge is ultimately founded in experience. This experience may be sensation or perception of the operation of our own mind. Knowledge, so Locke states, is the perception of the agreement or disagreement

of two ideas. The next step was taken by Berkeley. If actually all our knowledge is derived from perception, there is no meaning in the statement that the things really exist; because if the perception is given it cannot possibly make any difference whether the things exist or do not exist. Therefore, to be perceived is identical with existence. This line of argument then was extended to an extreme skepticism by Hume, who denied induction and causation and thereby arrived at a conclusion which if taken seriously would destroy the basis of all empirical science.

The criticism of metaphysical realism which has been expressed in empiristic philosophy is certainly justified in so far as it is a warning against the naïve use of the term "existence." The positive statements of this philosophy can be criticized on similar lines. Our perceptions are not primarily bundles of colors or sounds; what we perceive is already perceived as something, the accent here being on the word "thing," and therefore it is doubtful whether we gain anything by taking the perceptions instead of the things as the ultimate elements of reality.

The underlying difficulty has been clearly recognized by modern positivism. This line of thought expresses criticism against the naïve use of certain terms like "thing," "perception," "existence" by the general postulate that the question whether a given sentence has any meaning at all should always be thoroughly and critically examined. This postulate and its underlying attitude are derived from mathematical logic. The procedure of natural science is pictured as an attachment of symbols to the phenomena. The symbols can, as in mathematics, be combined according to certain rules, and in this way statements about the phenomena can be represented by combinations of symbols. However, a combination of symbols that does not comply with the rules is not wrong but conveys no meaning.

The obvious difficulty in this argument is the lack of any general criterion as to when a sentence should be considered as meaningless. A definite decision is possible only when the sentence belongs to a closed system of concepts and axioms, which in the development of natural science will be rather the exception than the rule. In some cases the conjecture that a certain sentence is meaningless has historically led to important progress, for it opened the way to the establishment of new connections which would have been impossible if the sentence had a meaning. An example in quantum theory that has already been discussed is the sentence: "In which orbit does the electron move around the nucleus?" But generally the positivistic scheme taken from mathematical logic is too narrow in a description of nature which necessarily uses words and concepts that are only vaguely defined.

The philosophic thesis that all knowledge is ultimately founded in experience has in the end led to a postulate concerning the logical clarification of any statement about nature. Such a postulate may have seemed justified in the period of classical physics, but since quantum theory we have learned that it cannot be fulfilled. The words "position" and "velocity" of an electron, for instance, seemed perfectly well defined as to both their meaning and their possible connections, and in fact they were clearly defined concepts within the mathematical framework of Newtonian mechanics. But actually they were not well defined, as is seen from the relations of uncertainty. One may say that regarding their position in Newtonian mechanics they were well defined, but in their relation to nature they were not. This shows that we can never know beforehand which limitations will be put on the applicability of certain concepts by the extension of our knowledge into the remote parts of nature, into which we

can only penetrate with the most elaborate tools. Therefore, in the process of penetration we are bound sometimes to use our concepts in a way which is not justified and which carries no meaning. Insistence on the postulate of complete logical clarification would make science impossible. We are reminded here by modern physics of the old wisdom that the one who insists on never uttering an error must remain silent.

A combination of those two lines of thought that started from Descartes, on the one side, and from Locke and Berkeley, on the other, was attempted in the philosophy of Kant, who was the founder of German idealism. That part of his work which is important in comparison with the results of modern physics is contained in *The Critique of Pure Reason*. He takes up the question whether knowledge is only founded in experience or can come from other sources, and he arrives at the conclusion that our knowledge is in part "a priori" and not inferred inductively from experience. Therefore, he distinguishes between "empirical" knowledge and knowledge that is "a priori." At the same time he distinguishes between "analytic" and "synthetic" propositions. Analytic propositions follow simply from logic, and their denial would lead to self-contradiction. Propositions that are not "analytic" are called "synthetic."

What is, according to Kant, the criterion for knowledge being "a priori"? Kant agrees that all knowledge starts with experience but he adds that it is not always derived from experience. It is true that experience teaches us that a certain thing has such or such properties, but it does not teach us that it could not be different. Therefore, if a proposition is thought together with its necessity it must be "a priori." Experience never gives to its judgments complete generality. For instance, the sentence "The sun rises every morning" means that we know no exception to

this rule in the past and that we expect it to hold in future. But we can imagine exceptions to the rule. If a judgment is stated with complete generality, therefore, if it is impossible to imagine any exception, it must be "a priori." An analytic judgment is always "a priori"; even if a child learns arithmetic from playing with marbles, he need not later go back to experience to know that "two and two are four." Empirical knowledge, on the other hand, is synthetic.

But are synthetic judgments a priori possible? Kant tries to prove this by giving examples in which the above criteria seem to be fulfilled. Space and time are, he says, a priori forms of pure intuition. In the case of space he gives the following metaphysical arguments:

1. Space is not an empirical concept, abstracted from other experiences, for space is presupposed in referring sensations to something external, and external experience is only possible through the presentation of space.

2. Space is a necessary presentation a priori, which underlies all external perceptions; for we cannot imagine that there should be no space, although we can imagine that there should be nothing in space.

3. Space is not a discursive or general concept of the relations of things in general, for there is only one space, of which what we call "spaces" are parts, not instances.

4. Space is presented as an infinite given magnitude, which holds within itself all the parts of space; this relation is different from that of a concept to its instances, and therefore space is not a concept but a form of intuition.

These arguments shall not be discussed here. They are mentioned merely as examples for the general type of proof that Kant has in mind for the synthetic judgments a priori.

With regard to physics Kant took as a priori, besides space and time, the law of causality and the concept of substance. In a later stage of his work he tried to include the law of conservation of matter, the equality of "actio and reactio" and even the law of gravitation. No physicist would be willing to follow Kant here, if the term "a priori" is used in the absolute sense that was given to it by Kant. In mathematics Kant took Euclidean geometry as "a priori."

Before we compare these doctrines of Kant with the results of modern physics we must mention another part of his work, to which we will have to refer later. The disagreeable question whether "the things really exist," which had given rise to empiristic philosophy, occurred also in Kant's system. But Kant has not followed the line of Berkeley and Hume, though that would have been logically consistent. He kept the notion of the "thing-in-itself" as different from the percept, and in this way kept some connection with realism.

Coming now to the comparison of Kant's doctrines with modern physics, it looks in the first moment as though his central concept of the "synthetic judgments a priori" had been completely annihilated by the discoveries of our century. The theory of relativity has changed our views on space and time, it has in fact revealed entirely new features of space and time, of which nothing is seen in Kant's a priori forms of pure intuition. The law of causality is no longer applied in quantum theory and the law of conservation of matter is no longer true for the elementary particles. Obviously Kant could not have foreseen the new discoveries, but since he was convinced that his concepts would be "the basis of any future metaphysics that can be called science" it is interesting to see where his arguments have been wrong.

As example we take the law of causality. Kant says that whenever we observe an event we assume that there is a foregoing event from which the other event must follow according to some rule. This is, as Kant states, the basis of all scientific work. In this discussion it is not important whether or not we can always find the foregoing event from which the other one followed. Actually we can find it in many cases. But even if we cannot, nothing can prevent us from asking what this foregoing event might have been and to look for it. Therefore, the law of causality is reduced to the method of scientific research; it is the condition which makes science possible. Since we actually apply this method, the law of causality is "a priori" and is not derived from experience.

Is this true in atomic physics? Let us consider a radium atom, which can emit an α -particle. The time for the emission of the α -particle cannot be predicted. We can only say that in the average the emission will take place in about two thousand years. Therefore, when we observe the emission we do not actually look for a foregoing event from which the emission must according to a rule follow. Logically it would be quite possible to look for such a foregoing event, and we need not be discouraged by the fact that hitherto none has been found. But why has the scientific method actually changed in this very fundamental question since Kant?

Two possible answers can be given to that question. The one is: We have been convinced by experience that the laws of quantum theory are correct and, if they are, we know that a foregoing event as cause for the emission at a given time cannot be found. The other answer is: We know the foregoing event, but not quite accurately. We know the forces in the atomic nucleus that are responsible for the emission of the α -particle.

But this knowledge contains the uncertainty which is brought about by the interaction between the nucleus and the rest of the world. If we wanted to know why the α -particle was emitted at that particular time we would have to know the microscopic structure of the whole world including ourselves, and that is impossible. Therefore, Kant's arguments for the a priori character of the law of causality no longer apply.

A similar discussion could be given on the a priori character of space and time as forms of intuition. The result would be the same. The a priori concepts which Kant considered an undisputable truth are no longer contained in the scientific system of modern physics.

Still they form an essential part of this system in a somewhat different sense. In the discussion of the Copenhagen interpretation of quantum theory it has been emphasized that we use the classical concepts in describing our experimental equipment and more generally in describing that part of the world which does not belong to the object of the experiment. The use of these concepts, including space, time and causality, is in fact the condition for observing atomic events and is, in this sense of the word, "a priori." What Kant had not foreseen was that these a priori concepts can be the conditions for science and at the same time can have only a limited range of applicability. When we make an experiment we have to assume a causal chain of events that leads from the atomic event through the apparatus finally to the eye of the observer; if this causal chain was not assumed, nothing could be known about the atomic event. Still we must keep in mind that classical physics and causality have only a limited range of applicability. It was the fundamental paradox of quantum theory that could not be foreseen by Kant. Modern physics has changed Kant's statement about the possibility of synthetic judgments a priori from a metaphysical one into a

practical one. The synthetic judgments a priori thereby have the character of a relative truth.

If one reinterprets the Kantian "a priori" in this way, there is no reason to consider the perceptions rather than the things as given. Just as in classical physics, we can speak about those events that are not observed in the same manner as about those that are observed. Therefore, practical realism is a natural part of the reinterpretation. Considering the Kantian "thing-in-itself" Kant had pointed out that we cannot conclude anything from the perception about the "thing-in-itself." This statement has, as Weizsäcker has noticed, its formal analogy in the fact that in spite of the use of the classical concepts in all the experiments a nonclassical behavior of the atomic objects is possible. The "thing-in-itself" is for the atomic physicist, if he uses this concept at all, finally a mathematical structure; but this structure is—contrary to Kant—indirectly deduced from experience.

In this reinterpretation the Kantian "a priori" is indirectly connected with experience in so far as it has been formed through the development of the human mind in a very distant past. Following this argument the biologist Lorentz has once compared the "a priori" concepts with forms of behavior that in animals are called "inherited or innate schemes." It is in fact quite plausible that for certain primitive animals space and time are different from what Kant calls our "pure intuition" of space and time. The latter may belong to the species "man," but not to the world as independent of men. But we are perhaps entering into too hypothetical discussions by following this biological comment on the "a priori." It was mentioned here merely as an example of how the term "relative truth" in connection with the Kantian "a priori" can possibly be interpreted.

Modern physics has been used here as an example or, we may say, as a model to check the results of some important philo-

sophic systems of the past, which of course were meant to hold in a much wider field. What we have learned especially from the discussion of the philosophies of Descartes and Kant may perhaps be stated in the following way:

Any concepts or words which have been formed in the past through the interplay between the world and ourselves are not really sharply defined with respect to their meaning; that is to say, we do not know exactly how far they will help us in finding our way in the world. We often know that they can be applied to a wide range of inner or outer experience, but we practically never know precisely the limits of their applicability. This is true even of the simplest and most general concepts like "existence" and "space and time." Therefore, it will never be possible by pure reason to arrive at some absolute truth.

The concepts may, however, be sharply defined with regard to their connections. This is actually the fact when the concepts become a part of a system of axioms and definitions which can be expressed consistently by a mathematical scheme. Such a group of connected concepts may be applicable to a wide field of experience and will help us to find our way in this field. But the limits of the applicability will in general not be known, at least not completely.

Even if we realize that the meaning of a concept is never defined with absolute precision, some concepts form an integral part of scientific methods, since they represent for the time being the final result of the development of human thought in the past, even in a very remote past; they may even be inherited and are in any case the indispensable tools for doing scientific work in our time. In this sense they can be practically *a priori*. But further limitations of their applicability may be found in the future.